

Fundamental of Remote Sensing

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Specification: Remote sensing and Applications

Practices

1. Exploring satellite data

2. Practice to measure the reflectance of surface object in the lab

3. Midterm presentation

4. Image pre-processing

5. Extract Information from satellite data

5. Extract Information from satellite data

Data: **Sentinel 2: S2A_MSIL2A_20211219T033141_N0301_R018_T48QWJ_20211219T055114**

Vector: **Hanoi.shp**

Tools: **SNAP**

Location: **Hanoi area**

1. Preprocessing image
2. Calculate index image: NDVI, NDWI, SATVI and classify the index images
2. Classification image
3. Accuracy assessment
4. Land cover change analysis

5.1 Preprocessing images

1/ Enhancement: *Color manipulation*

2/ Rescale the pixel size: *Raster/ Geometric/ Resampling*

3/ Clip image according to a specific area:

Import shape file: *Vector/Import*

Clip image: *Raster/Mask/Land_sea mask*

5.2 Calculate index image: NDVI, NDWI, SATVI

Tool: in QGIS; Raster/raster calculator

$$NDVI = \frac{(NIR - RED)}{(NIR + RED)}$$

Soil-Adjusted Total Vegetation Index

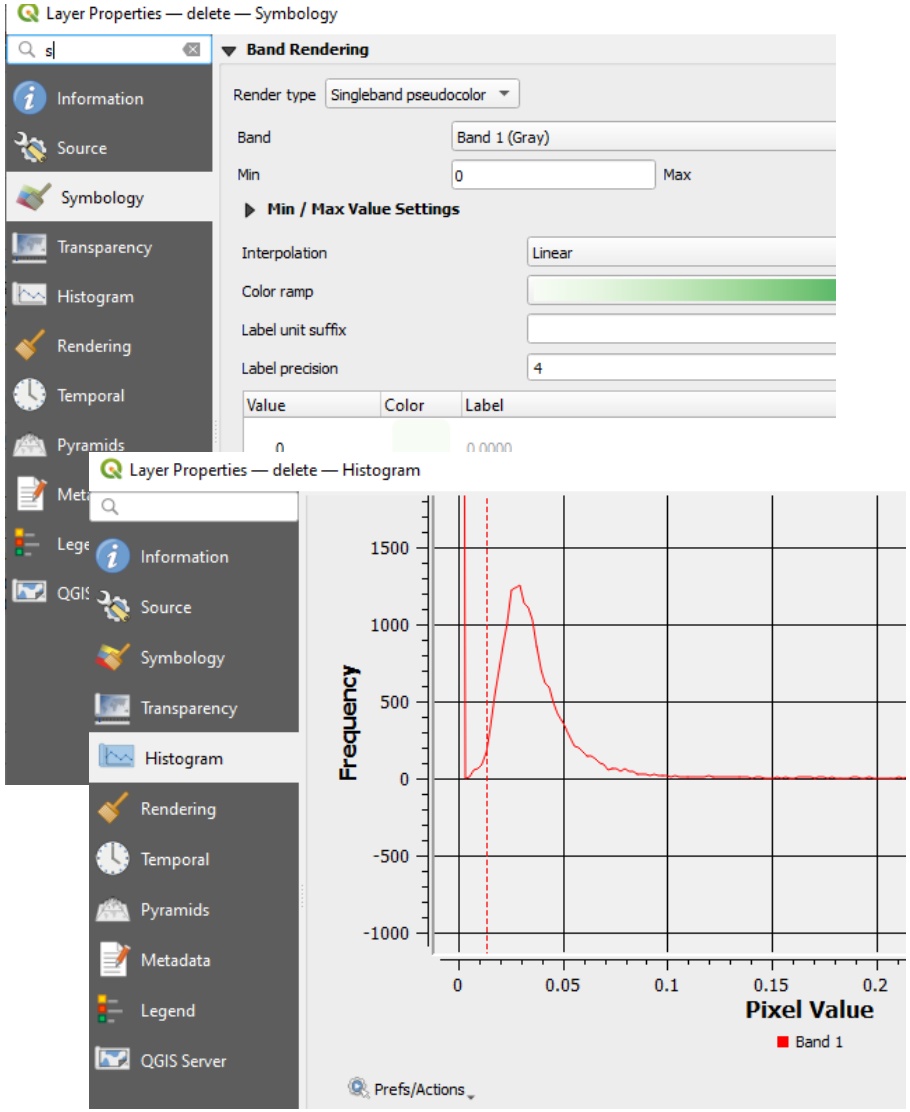
$$SATVI = \frac{SWIR1 - RED}{SWIR1 + RED + L} * (1 + L) - \frac{SWIR2}{2}$$

L is constant (related to the slope of the soil-line in a feature-space plot) that is usually set to 0.5

NDWI = (Green-SWIR1)/(SWIR1+Green)

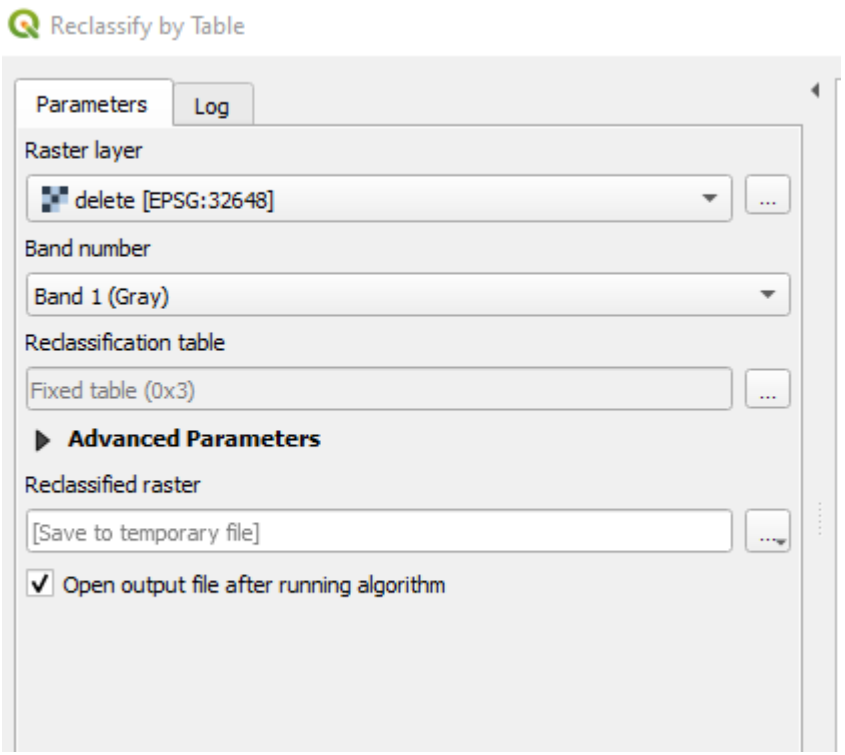
Slicing Index image into layer

1. Investigate the distribution of index image to define the thresholds



2/ Classifying the index image

Processing tools/ Raster analysis/ Reclassify by table



5.3 Image classification

Tools: A/ Unsupervised Classification:

B/ Supervised Classification

Steps: Build a classification system (for a land cover)

Collect training areas

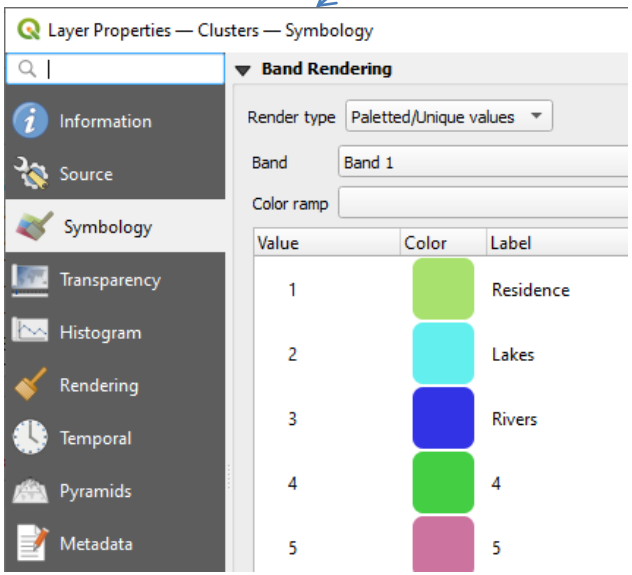
Analyze the spectral statistics of samples

Do classification with some techniques

A/ Unsupervised Classification with QGIS

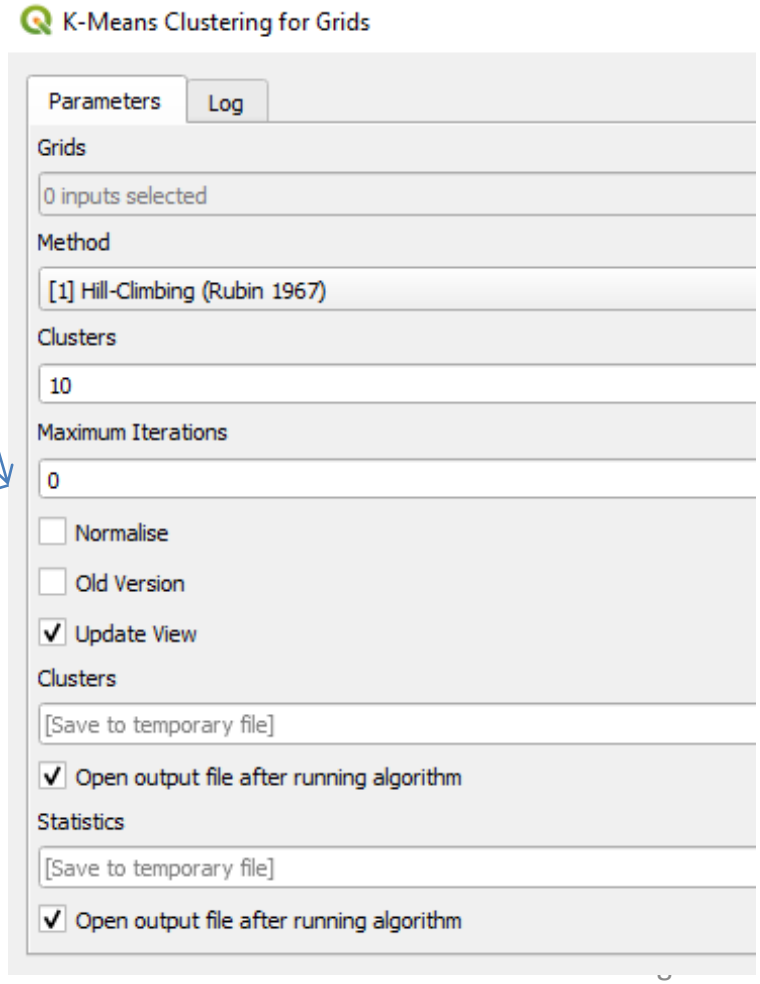
Develop a Classification system

Code	Class name	Description	Samples
1	Cloud	….	…
2	Cloud shadow	…	
3	Clear water	..	
4	Turbid water		
5	Residence		
6	Agriculture land		
7	Industrial area		
8	Bare land		

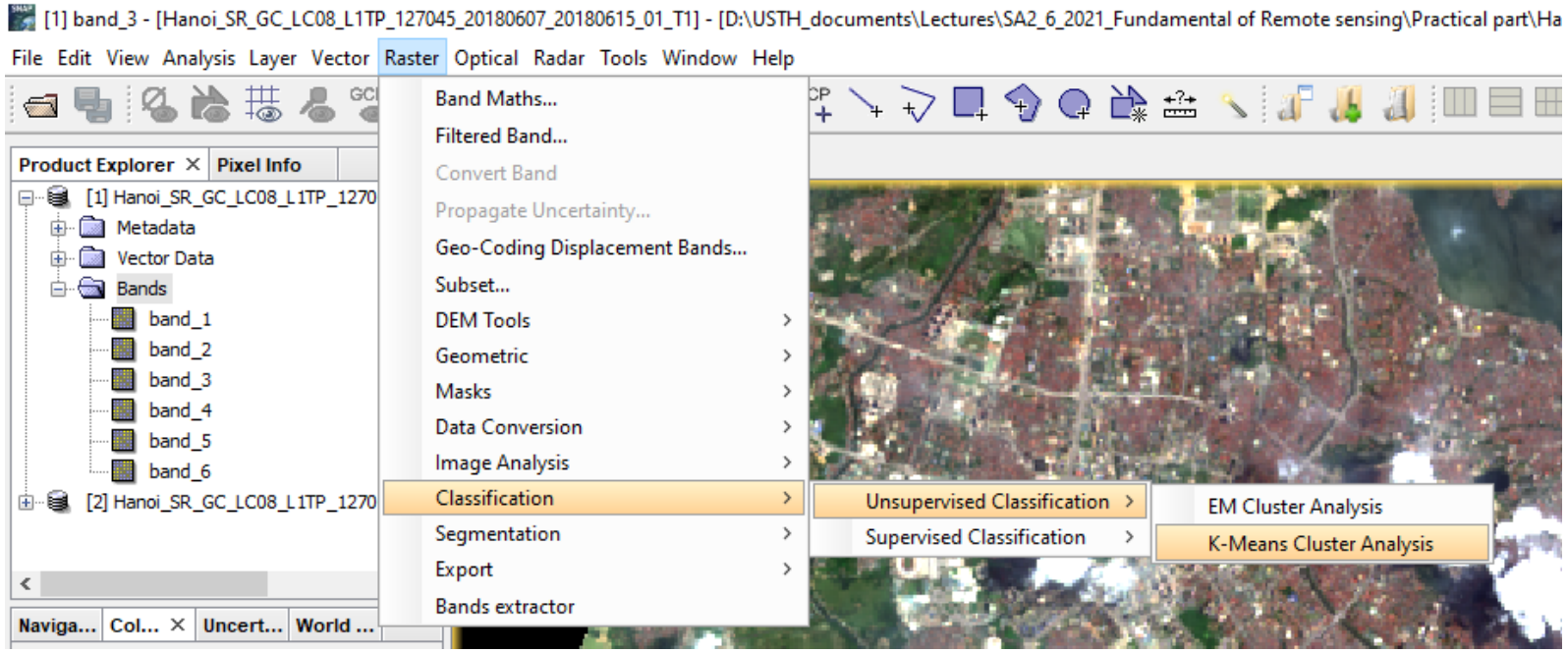


Naming classes

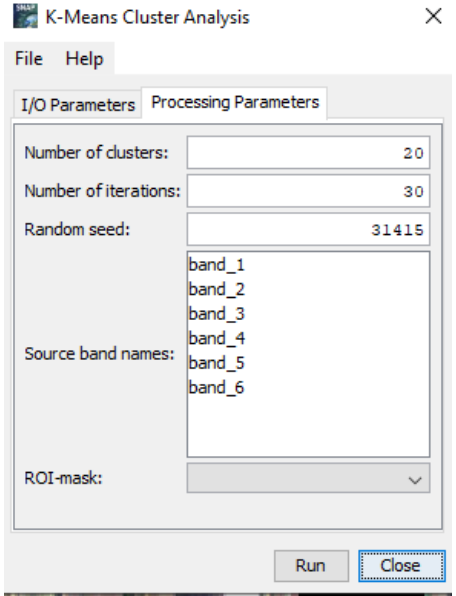
Processing Tool/ SAGA/ Image Analysis/ K-mean clustering for grid



A/ Unsupervised Classification with SNAP



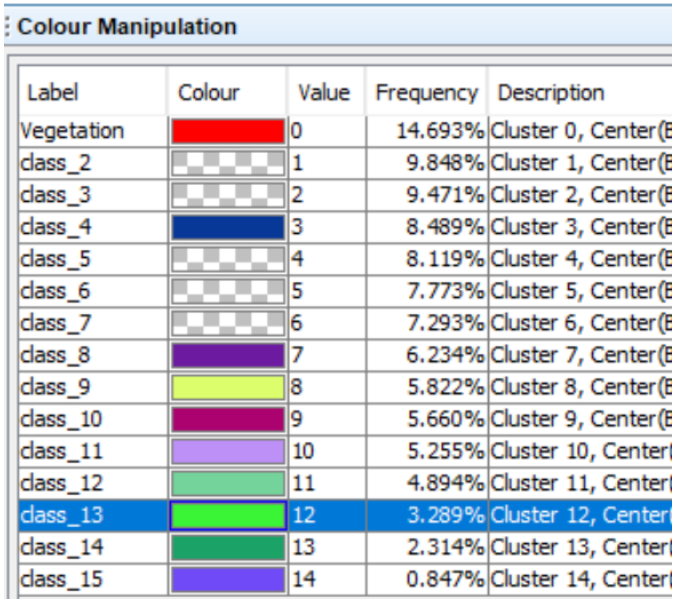
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A/ Unsupervised Classification with SNAP (Post-processing)

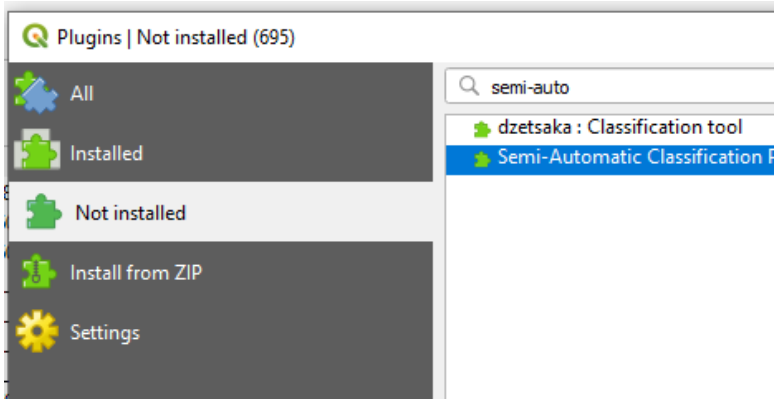
Code	Class name	Description	Samples
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View/ Tool windows/Color manipulation
To change color, Lables..

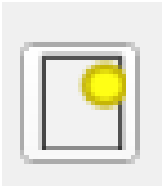


B/ Supervised Classification with QGIS

Install plugin:
Plugin/ Manage and install..



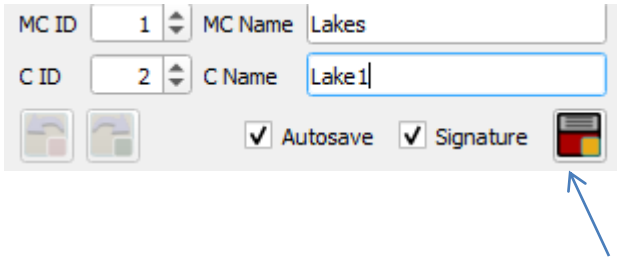
Create new training input



Create a training polygon



Assign attribute for training polygon



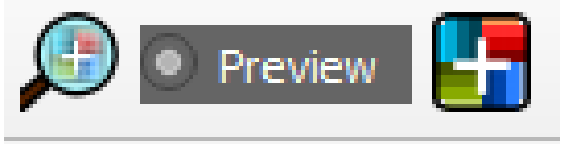
Sample statistic calculation

B/ Supervised Classification with QGIS

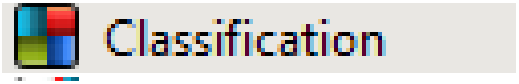
Observe Spectral plot of samples



Preview classification



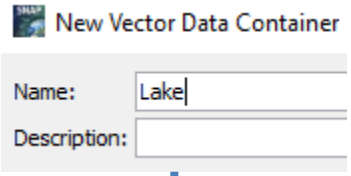
Do classifications



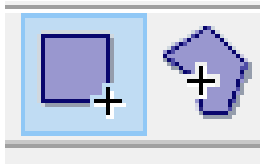
B/ Supervised Classification with SNAP

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Vector/ New vector data..



Drawing samples



Checking the quality of samples:
Analysis: **Scatter plot, Histogram, Statistic**

1/ Rater/ Classification/ Supervised classification/ Maximum Likelihood Classifier

2/ Rater/ Classification/ Supervised classification/ Random Forest Classifier

Be careful: Convert coordinate of image to Lat, Long to run the classification (Raster/ Geometric/ Reprojetion)

Change Colors of classes in Color manipulation of RF then Save Color palette. Import Color palette to ML to compare results from classifications

5.3 Land cover change analysis with QGIS/ SNAP

- Using Raster calculator (band math) tool to define the change area.

Classification with Minimum distance = land cover 2018

Classification with Minimum distance = land cover 2021

*Change map = Land cover 2018 *10 +Land cover 2021*

5.5 Accuracy assessment in QGIS

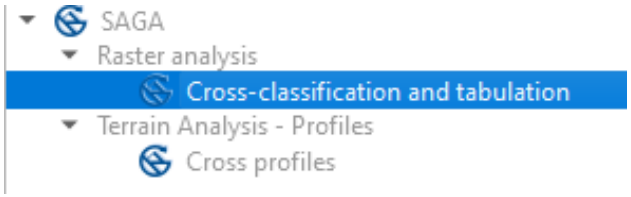
Assuming that:

Classification with Maximum likelihood = land cover 2021

Classification with Random Forest = Reference data

QGIS: Processing toolbox/ SAGA/ Raster analysis/ Cross-Classification..





Export results to Excel file. Then convert to % to see the accuracy of classification

SNAP: *Confuse map = RF*10 +ML*

Color Manipulation: See the % of confuse

Thank you for your attention!